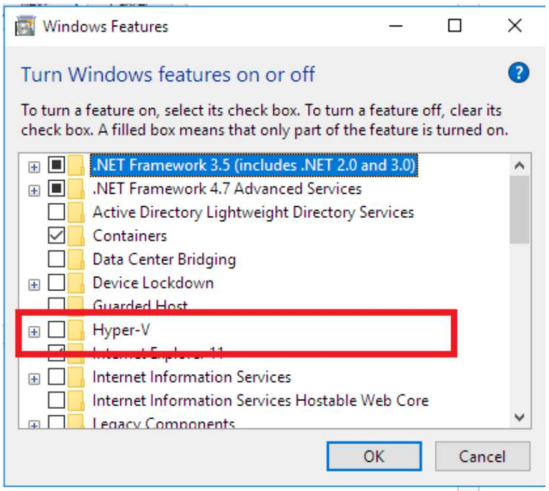
In this document students will learn about installing a Linux VM.
Steps to achieve a functional VM:

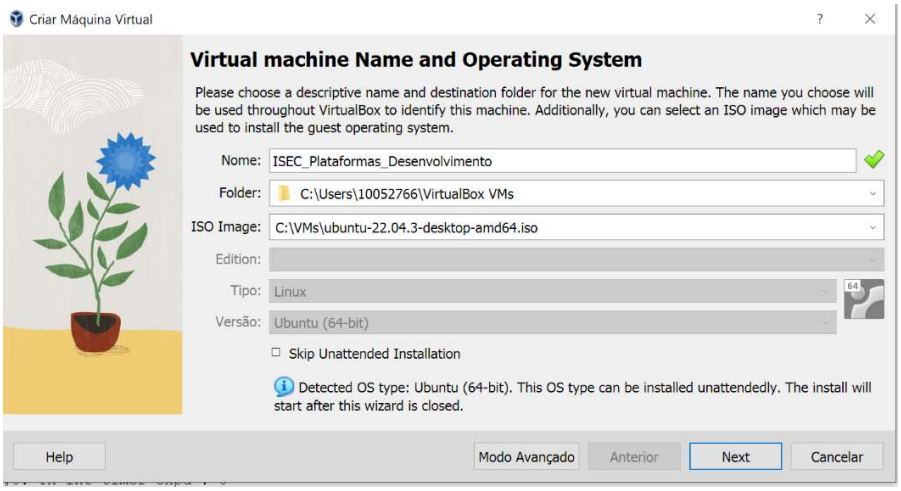
1. Install Virtualbox to manage VMs:
    a. https://www.virtualbox.org/wiki/Downloads

2. Download the Ubuntu Desktop Image
    a. URL for Ubuntu 22.04.3 LTS Desktop (but other version can be installed):
        i. https://releases.ubuntu.com/jammy/

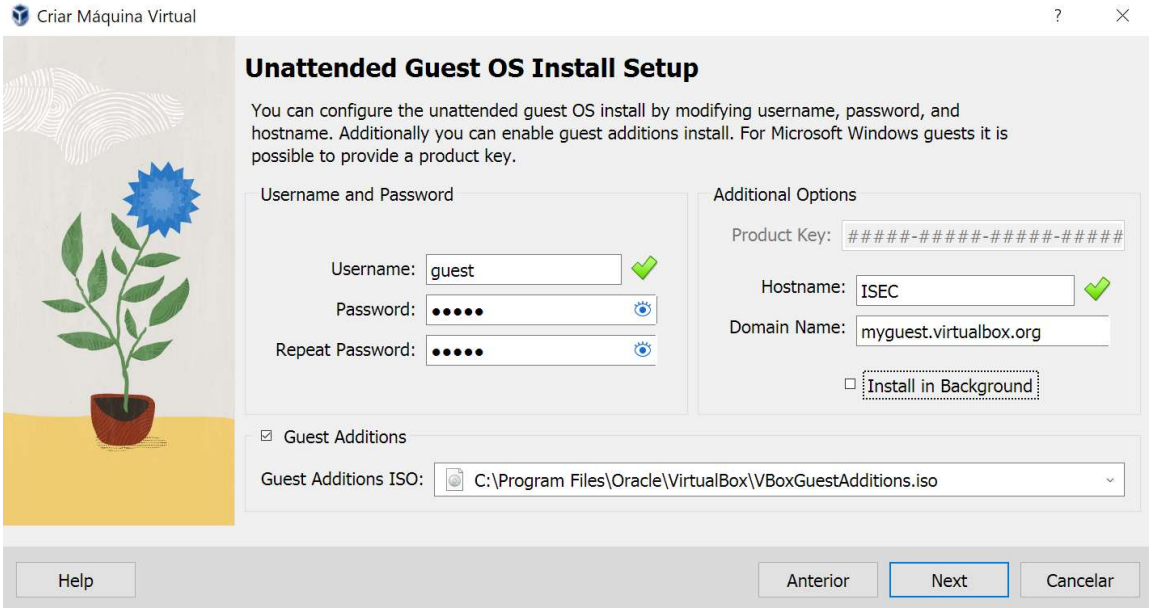
3. In Windows OS uncheck the Hyper-V functionality. Note that Hyper-V is normally needed for WSL processes.



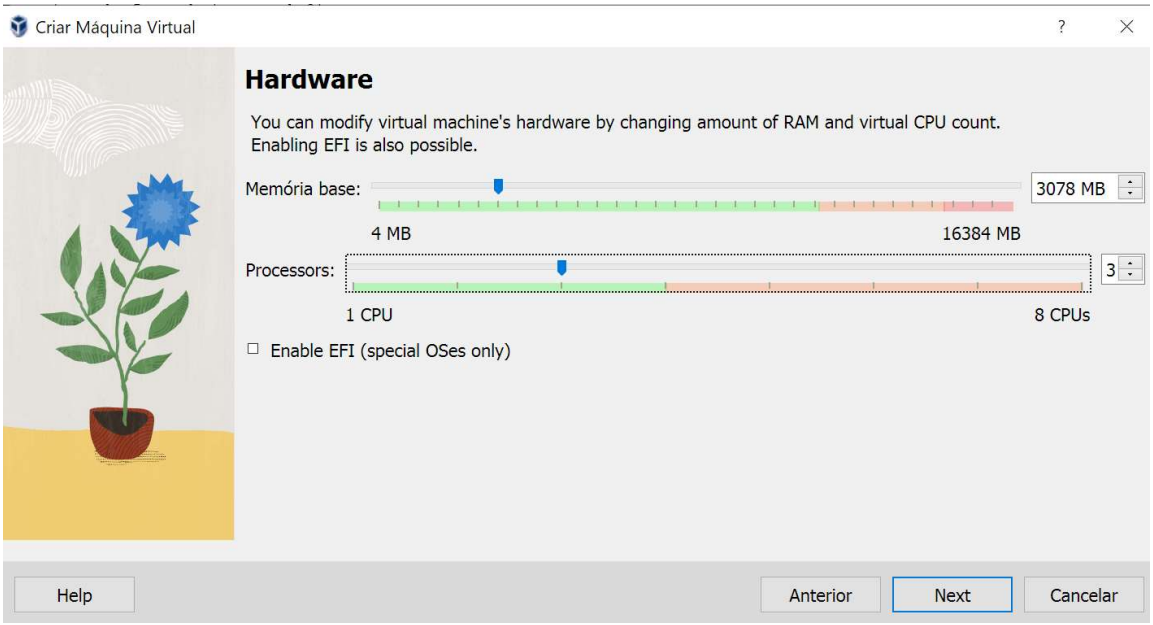
4. Install Ubuntu Image. For this, create a new VM with Ubuntu Image:



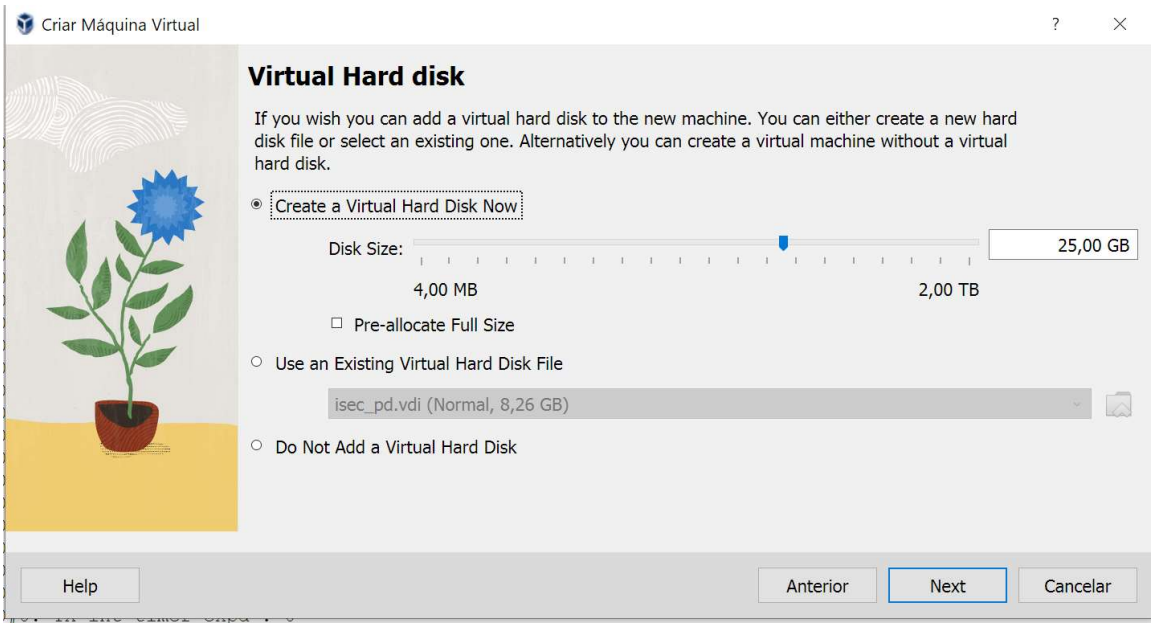
5. The password from image below will be needed to access root user. Ensure you save it.



6. Choose at least 3GB of RAM and 3 processors.

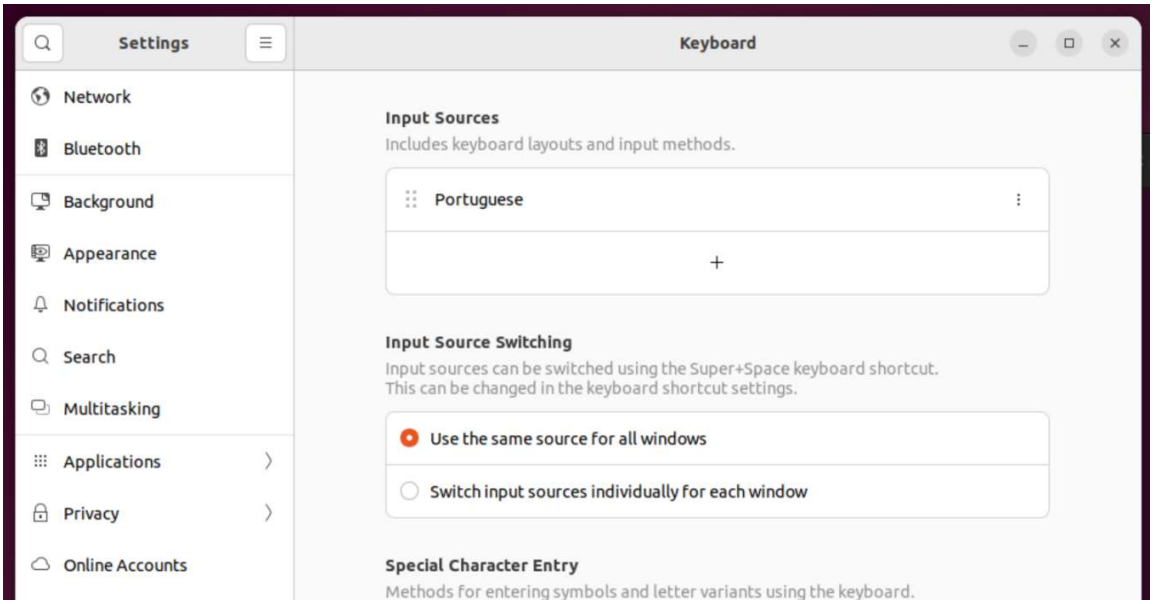


7. Create a new Virtual Hard Disk

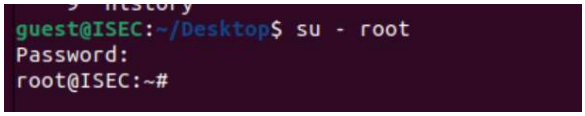


Afterwards, the installation may take a while, for downloading all the content needed by Ubuntu.

Note1: You can also install “Upgrade Guest Additions” to be able to access clipboard between host and VM.
Note2: You can change display resolution in “Virtual Display 1“ settings.
Note3: Ubuntu might install by default the UK keyboard. You can change it in Ubuntu settings:



8. You can access root escalation access using the password inserted in the Guest OS Install Setup. For this just open a new terminal and type “su - root” and then the password.

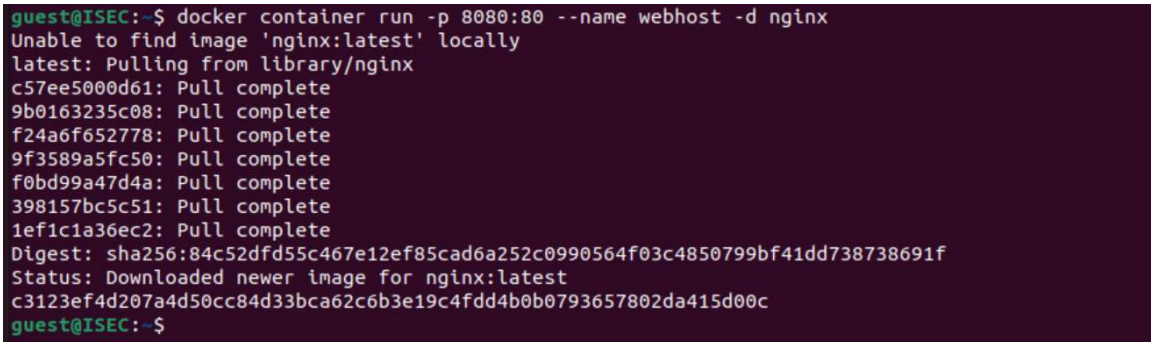


9. We are going now to give sudo privileges to the guest user (or equivalent user credentials from “Guest OS Install Setup”). For this, in root account just add guest user to sudo using the command “usermod -aG sudo guest”.

10. Install docker in the Ubuntu VM. For this, in a **new** terminal with guest user (or user created in this tutorial) type the command:

    $sudo apt-get install curl
    $curl -sSL https://get.docker.com/ | sh
    $sudo usermod -aG docker guest

If everything went well, try running the following docker command. The log should be similar to below image.



Hint: What happens when you go to URL “localhost:8080” inside the VM?

11. Create DockerHub account and private repository

Create a new account on website https://hub.docker.com/
This account will be used to send new images to private or public repositories owned by the user on dockerhub. Hence, these new created images will be available to pull from dockerhub to other locations than only the student laptop.

Try also to create private repositories, to have more control over the created images.
Below image shows the creation of a private repository and how to push new images to that repository.

